

[Courseware \(/courses/HumanitiesScience/StatLearning/Winter2014/courseware\)](/courses/HumanitiesScience/StatLearning/Winter2014/courseware)[Course Info \(/courses/HumanitiesScience/StatLearning/Winter2014/info\)](/courses/HumanitiesScience/StatLearning/Winter2014/info)[Discussion \(/courses/HumanitiesScience/StatLearning/Winter2014/discussion/forum\)](/courses/HumanitiesScience/StatLearning/Winter2014/discussion/forum)[Wiki \(/courses/HumanitiesScience/StatLearning/Winter2014/course_wiki\)](/courses/HumanitiesScience/StatLearning/Winter2014/course_wiki)[Progress \(/courses/HumanitiesScience/StatLearning/Winter2014/progress\)](/courses/HumanitiesScience/StatLearning/Winter2014/progress)[Lecture Slides \(pdf\) \(/courses/HumanitiesScience/StatLearning/Winter2014/ba1951b8f66c4cdca2fcaabcdc91b792/\)](/courses/HumanitiesScience/StatLearning/Winter2014/ba1951b8f66c4cdca2fcaabcdc91b792/)[R Sessions \(/courses/HumanitiesScience/StatLearning/Winter2014/0d68641c19484ef9aa6aeea03426dc68/\)](/courses/HumanitiesScience/StatLearning/Winter2014/0d68641c19484ef9aa6aeea03426dc68/)

10.R.1 (1/1 point)

Suppose we want to fit a linear regression, but the number of variables is much larger than the number of observations. In some cases, we may improve the fit by reducing the dimension of the features before.

In this problem, we use a data set with $n = 300$ and $p = 200$, so we have more observations than variables, but not by much. Load the data `x`, `y`, `x.test`, and `y.test` from [10.R.RData \(/c4x/HumanitiesScience/StatLearning/asset/10.R.RData\)](/c4x/HumanitiesScience/StatLearning/asset/10.R.RData).

First, concatenate `x` and `x.test` using the `rbind` functions and perform a principal components analysis on the concatenated data frame. To within 10% relative error, what proportion of the variance is explained by the first five principal components?

Answer: 0.34986**EXPLANATION**

use `"vars = prcomp(rbind(x,x.test),scale=TRUE)$sdev^2"`

[Hide Answer](#)*You have used 3 of 5 submissions*

10.R.2 (1/1 point)

The previous answer suggests that a relatively small number of "latent variables" account for a substantial fraction of the features' variability. We might believe that these latent variables are more important than linear combinations of the features that have low variance.

We can try forgetting about the raw features and using the first five principal components (computed on `rbind(x,x.test)`) instead as low-dimensional derived features. What is the mean-squared test error if we regress `y` on the first five principal components, and use the resulting model to predict `y.test`?

Answer: 1.088

EXPLANATION

In the actual data generating model for this example, the features may be noisy proxies for a few latent variables that actually drive the response. This is not an uncommon situation when we have high-dimensional data.

Hide Answer

You have used 1 of 5 submissions

10.R.3 (1/1 point)

Now, try an OLS linear regression of y on the matrix x . What is the mean squared prediction error if we use the fitted model to predict y_{test} from x_{test} ?

3.657197

3.657197

Answer: 3.90714

EXPLANATION

The mean squared error is worse because of the variance involved in fitting a very high dimensional model. As it turned out here, the large-variance directions of x turned out to be the important ones for predicting y . Note that this need not always be the case, but it often is.

Hide Answer

You have used 1 of 5 submissions



[Terms of Service \(/tos\)](#) [Privacy Policy \(/tos#privacy\)](#) [Honor Code \(/tos#honor\)](#)
[Copyright \(/tos#copyright\)](#) [Careers \(/about#careers\)](#) [Contact \(/about#contact\)](#)
[Help \(/faq\)](#)

Built on [OpenEdX \(http://code.edx.org\)](http://code.edx.org).


(<http://www.stanford.edu>)

[SU Home \(http://www.stanford.edu\)](http://www.stanford.edu)
[Maps & Directions \(http://visit.stanford.edu/plan/maps.html\)](http://visit.stanford.edu/plan/maps.html)
[Search Stanford \(http://www.stanford.edu/search/\)](http://www.stanford.edu/search/)
[Terms of Use \(http://www.stanford.edu/site/terms.html\)](http://www.stanford.edu/site/terms.html)
[Copyright Complaints \(http://www.stanford.edu/site/copyright.html\)](http://www.stanford.edu/site/copyright.html)

© Stanford University, Stanford, California 94305